

AP CSP | Lesson 24: Packets & Routing

Homework | Multiple-choice and written practice

Name: _____ Class: _____ Date: _____

Time: 45 minutes | Total: 30 teacher marks (not an AP score) | Show working.

Original AP-style MCQs and classroom written practice; these are not official AP questions.

Q5 requires exactly TWO choices. Show paths, calculations and reasons. Treat all networks and timings as the stated classroom models.

1. What is the packet payload? [1 mark]

- A. Its destination address
- B. The data chunk it carries
- C. Its route-count total
- D. Its complete protocol specification

Answer: _____

2. Packets with IDs 4, 2, 1, 3 arrive. Which order reconstructs the original message? [1 mark]

- A. 4, 2, 1, 3
- B. 3, 1, 2, 4
- C. 1, 2, 3, 4
- D. Any order gives the same text

Answer: _____

3. Packet 3 never arrives, but 1, 2 and 4 do. What should the classroom receiver report? [1 mark]

- A. Incomplete buffer
- B. Successful complete transfer
- C. An empty buffer
- D. Packet 4 replaces packet 3

Answer: _____

4. What does IP best-effort service imply? [1 mark]

- A. All packets arrive in order.
- B. Only one route is possible.
- C. Lost packets are always replaced by IP.
- D. Delivery is attempted without guaranteeing arrival.

Answer: _____

5. Which TWO statements are correct? [1 mark]

- A. TCP supports reliable ordered byte-stream delivery.
- B. TCP makes permanent network failure impossible.
- C. UDP itself does not guarantee delivery or ordering.
- D. All packets of a message must use one fixed route.

Answer: _____

6. Why might an application skip an old live update rather than wait for recovery? [1 mark]

- A. All lost data are harmless.
- B. A new update may be more useful than the stale one.
- C. UDP always has lower total delay.
- D. The Internet cannot retransmit.

Answer: _____

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Written responses | Questions 7-9

7. [4 marks]

Split HELLO AP CSP into chunks of 3 characters. (a) Give packet 1. (b) Give packet 2. (c) Give packet 3. (d) Give packet 4. Preserve spaces inside quotes.

8. [4 marks]

Arrival order is 4, 2, 1, 3. (a) Which IDs are present after two arrivals? (b) Which are missing then? (c) Give final reconstruction order. (d) Explain why arrival order is insufficient.

9. [4 marks]

Only IDs 1,2,4 arrive. (a) State the buffer status. (b) Identify the absent chunk. (c) Describe a recovery action. (d) State what evidence is needed before claiming completion.

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Written responses | Questions 10-12

10. [4 marks]

The original route is S-A-C-T. A-C fails; A-D and D-T remain. (a) Give an alternative path. (b) Count its hops. (c) Does the destination necessarily change? (d) Why does this not prove all packets arrive?

11. [4 marks]

Compare TCP and UDP. (a) State TCP's application-facing service. (b) Name one TCP recovery mechanism. (c) State a guarantee UDP itself lacks. (d) Explain why TCP does not make every transfer succeed.

12. [4 marks]

An app receives frequent location updates. (a) Give a reason to prefer fresh data over a missing old update. (b) Give a task needing exact ordered data instead. (c) Why is "UDP is always faster" unjustified? (d) Name one limitation of our manual-retry model.
